

Delta Sets as Formal Boundary Objects in Incomplete Theories

A Formal Bridge Between Incompleteness and Structural Boundaries

Abstract

We introduce *Delta sets* (Δ) as formally defined regions of non-derivability within a theory. We prove that in incomplete theories, Δ is non-empty and induces a natural boundary structure. This establishes a precise link between logical incompleteness and boundary-based formulations used in the TNA framework, eliminating the need for analogical interpretation.

1. Preliminaries

Definition 1 (Formal Theory)

$$T = (\mathcal{L}, A, R)$$

with:

- $\mathcal{L}$: formal language
- A : axioms
- R : inference rules

Definition 2 (Derivability)

$$T \vdash \varphi$$

denotes that φ is derivable from T .

Definition 3 (Decidability over Domain Ω)

Let $\Omega \subseteq \mathcal{L}$.

A theory T is complete over Ω if:

$$\forall \varphi \in \Omega, T \vdash \varphi \vee T \vdash \neg \varphi$$

2. Delta Set Definition

Definition 4 (Delta Set)

$$\Delta(T, \Omega) = \{\varphi \in \Omega \mid T \not\vdash \varphi \wedge T \not\vdash \neg \varphi\}$$

Interpretation

Δ is the set of:

undecidable propositions relative to T within Ω

3. Fundamental Result

Theorem 1 (Existence of Delta Sets)

If T is incomplete over Ω , then:

$$\Delta(T, \Omega) \neq \emptyset$$

Proof

By definition of incompleteness, there exists $\varphi \in \Omega$ such that:

$$T \not\vdash \varphi \wedge T \not\vdash \neg \varphi$$

Thus:

$$\varphi \in \Delta(T, \Omega)$$

Hence:

$$\Delta(T, \Omega) \neq \emptyset$$

■

4. Boundary Interpretation

We now define a structural boundary.

Definition 5 (Boundary Set)

$$\partial\Omega_T = \Delta(T, \Omega)$$

Proposition 1

The boundary of a theory relative to Ω is exactly its Delta set.

Implication

$$\text{Incompleteness} \iff \partial\Omega_T \neq \emptyset$$

5. Structural Properties of Δ

Proposition 2 (Non-Closure)

Δ is not closed under derivation:

$$\varphi \in \Delta \not\Rightarrow T \vdash \varphi$$

Proposition 3 (Expansion Sensitivity)

If $T' = T \cup A'$, then:

$$\Delta(T', \Omega) \subseteq \Delta(T, \Omega)$$

provided A' resolves some undecidable propositions.

Proposition 4 (Persistence)

For sufficiently rich theories:

$$\Delta(T, \Omega) \text{ persists under finite extensions}$$

6. Delta-Induced Structure

Δ induces a partition of Ω :

$$\Omega = \Omega^+ \cup \Delta \cup \Omega^-$$

where:

- $\Omega^+ = \{\varphi \mid T \vdash \varphi\}$
 - $\Omega^- = \{\varphi \mid T \vdash \neg\varphi\}$
-

7. Connection to Boundary Dynamics

We now connect Δ to structural dynamics.

Definition (Local Failure Indicator)

Define:

$$R(\varphi) = \begin{cases} 1 & \text{if } T \vdash \varphi \text{ or } T \vdash \neg\varphi \\ 0 & \text{if } \varphi \in \Delta \end{cases}$$

Observation

$$\varphi \in \Delta \Rightarrow R(\varphi) = 0$$

Interpretation

Δ corresponds to:

regions of zero derivability reliability

8. From Δ to Boundary Pressure

Define:

$$F_B = \sum_{\varphi \in \Delta} w(\varphi)$$

where $w(\varphi)$ is a weighting function.

Minimal Form

$$F_B = |\Delta|$$

Extended Form

$$F_B = \sum_{\varphi \in \Delta} [\beta U(\varphi) + \gamma C(\varphi)]$$

9. Key Result

Theorem 2 (Delta-Pressure Link)

$$\Delta(T, \Omega) \neq \emptyset \Rightarrow F_B > 0$$

10. Consequence

A theory with Δ :

- cannot be structurally closed
 - contains intrinsic boundary regions
 - generates pressure for extension
-

11. Relation to Previous Work

- Formal paper $\rightarrow$ non-self-sufficiency
 - Interpretation $\rightarrow$ circularity $\rightarrow$ boundary
 - This paper $\rightarrow$ **explicit construction of boundary (Δ)**
-

12. Final Statement

$$\text{Incompleteness} \Rightarrow \text{Delta Set} \Rightarrow \text{Boundary}$$

13. Conclusion

We have shown that:

- Δ is a formally defined object
- it emerges directly from incompleteness
- it constitutes the boundary of a theory

Thus, the transition from axiomatic systems to boundary-based formulations is no longer analogical, but formally grounded.